

# CM12004: Problem Sheet

1. (i) 

| $X$ | $Y$ | $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-----|-----|-------------------|
| 0   | 0   | 0   | 1   | 1                 |
| 0   | 1   | 1   | 1   | 1                 |
| 1   | 0   | 1   | 1   | 1                 |
| 1   | 1   | 1   | 0   | 0                 |

**conclusion:** neither
- (ii) 

| $X$ | $Y$ | $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-----|-----|-------------------|
| 0   | 0   | 0   | 1   | 1                 |
| 0   | 1   | 1   | 0   | 0                 |
| 1   | 0   | 1   | 0   | 0                 |
| 1   | 1   | 1   | 0   | 0                 |

**conclusion:** neither
- (iii) 

| $X$ | $Y$ | $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-----|-----|-------------------|
| 0   | 0   | 1   | 1   | 1                 |
| 0   | 1   | 1   | 1   | 1                 |
| 1   | 0   | 0   | 0   | 1                 |
| 1   | 1   | 1   | 1   | 1                 |

**conclusion:** tautology
- (iv) 

| $X$ | $Y$ | $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-----|-----|-------------------|
| 0   | 0   | 1   | 1   | 1                 |
| 0   | 1   | 1   | 1   | 1                 |
| 1   | 0   | 1   | 1   | 1                 |
| 1   | 1   | 0   | 0   | 1                 |

**conclusion:** tautology
- (v) 

| $X$ | $Y$ | $Z$ | $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-----|-----|-----|-------------------|
| 0   | 0   | 0   | 0   | 0   | 1                 |
| 0   | 0   | 1   | 0   | 0   | 1                 |
| 0   | 1   | 0   | 0   | 0   | 1                 |
| 0   | 1   | 1   | 0   | 1   | 1                 |
| 1   | 0   | 0   | 0   | 1   | 1                 |
| 1   | 0   | 1   | 1   | 1   | 1                 |
| 1   | 1   | 0   | 1   | 1   | 1                 |
| 1   | 1   | 1   | 1   | 1   | 1                 |

**conclusion:** tautology
- (vi) 

| $X$ | $Y$ | $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-----|-----|-------------------|
| 0   | 0   | 1   | 1   | 1                 |
| 0   | 1   | 1   | 1   | 1                 |
| 1   | 0   | 0   | 1   | 1                 |
| 1   | 1   | 1   | 0   | 0                 |

**conclusion:** neither

|       |     |     |     |     |                   |                            |
|-------|-----|-----|-----|-----|-------------------|----------------------------|
|       | $X$ | $Y$ | $P$ | $Q$ | $P \rightarrow Q$ |                            |
|       | 0   | 0   | 1   | 0   | 0                 |                            |
| (vii) | 0   | 1   | 1   | 1   | 1                 | <b>conclusion:</b> neither |
|       | 1   | 0   | 0   | 0   | 1                 |                            |
|       | 1   | 1   | 1   | 0   | 0                 |                            |

|        |     |     |     |     |     |                   |                              |
|--------|-----|-----|-----|-----|-----|-------------------|------------------------------|
|        | $X$ | $Y$ | $Z$ | $P$ | $Q$ | $P \rightarrow Q$ |                              |
|        | 0   | 0   | 0   | 1   | 1   | 1                 |                              |
|        | 0   | 0   | 1   | 1   | 1   | 1                 |                              |
|        | 0   | 1   | 0   | 0   | 1   | 1                 |                              |
| (viii) | 0   | 1   | 0   | 0   | 1   | 1                 | <b>conclusion:</b> tautology |
|        | 0   | 1   | 1   | 1   | 1   | 1                 |                              |
|        | 1   | 0   | 0   | 0   | 0   | 1                 |                              |
|        | 1   | 0   | 1   | 0   | 1   | 1                 |                              |
|        | 1   | 1   | 1   | 0   | 0   | 1                 |                              |
|        | 1   | 1   | 1   | 1   | 1   | 1                 |                              |
|        | 1   | 1   | 1   | 1   | 1   | 1                 |                              |

2. (a) binary logical connectives operate on 2 binary inputs. each binary input can have 2 values, 1 or 0.

$\therefore$  there are 4 possible combinations of inputs that can be operated on by a connective.

$\implies$  each logical connective can have 4 possible binary outputs, one corresponding to each possible input combination.

having 4 possible binary outputs  $\implies$  there are 16 possible combinations of outputs a connective can have.

$\therefore$  **answer: there are 16 possible binary logical connectives.**

(b) (i)  $\neg(\neg(P \wedge \neg Q) \wedge \neg(\neg P \wedge Q))$

(ii)  $\neg(\neg P \vee Q) \vee \neg(P \vee \neg Q)$

(iii)  $(P \rightarrow Q) \rightarrow \neg(Q \rightarrow P)$

(c)

$$\bar{\wedge} = \neg(X \wedge Y) \quad (1)$$

$$\neg = X \bar{\wedge} X \quad (2)$$

$$\wedge = \neg(\neg(X \wedge Y)) = (X \bar{\wedge} Y) \bar{\wedge} (X \bar{\wedge} Y) \quad (3)$$

$$\vee = \neg(\neg X \wedge \neg Y) = (X \bar{\wedge} X) \bar{\wedge} (Y \bar{\wedge} Y) \quad (4)$$

$$\rightarrow = \neg X \vee Y = \neg(X \wedge \neg Y) = X \bar{\wedge} (Y \bar{\wedge} Y) \quad (5)$$

(d)

$$\bar{\vee} = \neg(X \vee Y) \quad (6)$$

$$\neg = X \bar{\vee} X \quad (7)$$

$$\vee = \neg(\neg(X \vee Y)) = (X \bar{\vee} Y) \bar{\vee} (X \bar{\vee} Y) \quad (8)$$

$$\wedge = \neg(\neg X \vee \neg Y) = (X \bar{\vee} X) \bar{\vee} (Y \bar{\vee} Y) \quad (9)$$

$$\rightarrow = \neg X \vee Y = ((X \bar{\vee} X) \bar{\vee} Y) \bar{\vee} ((X \bar{\vee} X) \bar{\vee} Y) \quad (10)$$

3. (a) (i) **question:**  $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} (x^2 < y + 1)$

**true**

*Proof.*

$$x^2 < y + 1 \quad (11)$$

$$\implies x^2 - 1 < y \quad (12)$$

$$\implies y > x^2 - 1 \quad (13)$$

if  $x \in \mathbb{Z}$  and  $y \in \mathbb{Z} \implies \exists y \mid y > x^2 - 1 \because \mathbb{Z}$  has no upper bound, so there will always be a bigger  $y \forall x$ .  $\square$

- (ii) **question:**  $\exists x \in \mathbb{Z} \forall y \in \mathbb{Z} (x^2 < y + 1)$

**false**

*Proof.*

$$x^2 < y + 1 \quad (14)$$

$$\implies x^2 - 1 < y \quad (15)$$

we know this to be false, as for it to be true,  $x^2 - 1$  would have to be smaller than every integer, which is impossible.  $\square$

- (iii) **question:**  $\exists x \in \mathbb{Z} \forall x \in \mathbb{Z} (x^2 < y + 1)$

**false**

*Proof.*

$$x^2 < y + 1 \quad (16)$$

$$\implies x^2 - 1 < y \quad (17)$$

for a similar reason to (ii) we know this to also be false, as for it to be true,  $y$  would have to be bigger than every integer, which is impossible.  $\square$

- (iv) **question:**  $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} ((x < y) \rightarrow (x^2 < y^2))$

**true**

*Proof.* let  $y = x + 1$ .

$x < y$  is true  $\forall x, y \because x < x + 1$

$\implies (x < y) \rightarrow (x^2 < y^2)$  is true if  $(x^2 < y^2)$  is true.

$$y^2 = (x + 1)^2 = x^2 + 2x + 1 \implies y^2 > x^2$$

$\therefore (x < y) \rightarrow (x^2 < y^2)$  is true when  $y = x + 1$ .

$\therefore \exists y \in \mathbb{Z} (y = x + 1) \forall x \in \mathbb{Z} ((x < y) \rightarrow (x^2 < y^2))$   $\square$

- (b) (i) let  $x = 6, y = 5. \implies x^2 = 36, y + 1 = 6. 36 > 6 \therefore$  **false.**

- (ii) let  $x = 2, y = 5. \implies x^2 = 4, y + 1 = 6. 4 < 6 \therefore$  **true.**

- (iii) let  $y = -1. \implies y + 1 = 0. \forall x \in \mathbb{Z}, x^2 \geq 0 \therefore \nexists x \mid x^2 < 0 \therefore$  **false.**

- (iv) let  $x = 0, \implies x^2 = 0$ . if  $y > 0, \implies y > x$  but also  $y^2 > 0 \therefore (x^2 < y^2)$  is true. if  $y \leq 0, \implies y < x \therefore ((x < y) \rightarrow (x^2 < y^2))$  is true.  $\therefore$  **true.**

4. (a) •  $2^\emptyset = 1$   
•  $2^{\{0\}} = 2$   
•  $2^{\{0\} \cup \{1\}} = 4$   
•  $2^{\{\emptyset, 0, 1\}} = 2^3 = 8$   
•  $2^{2^{2^{\{0, 1\}}}} = 2^{2^{2^2}} = 2^{16} = 65536$
- (b) (i) for  $A, |A| = 2^n$ . therefore, there are  $2^n$  values of  $S$ . for each value of  $s$ , either  $x \in S$ , or  $x \notin S$ .  $\implies$  there are  $2^{n-1}$  values of  $S$  where  $x \in S$ .  $x \in A$ ,  $\therefore$  there are  $n$  values of  $x$ .  $\implies$  there are  $n \cdot 2^{n-1}$  pairs  $\{(x, S) \mid x \in S, \S \in 2^A\}$ .
- (ii) for each element  $x \in A$ , there are three options. either:  $x \in S, x \in T$ , or  $x \notin S \cup T$ .  $\implies$ , for each element  $\{1, 2, \dots, n\}$ , it can be in either one of the two sets  $S, T$ , or neither.  $\therefore$  there are three possibilities for each item. as there are  $|A| = n$  elements,  $\implies$  the total number of ways to form pairs  $\{(S, T) \mid S \in 2^A, T \in 2^A, S \cap T = \emptyset\}$  is  $3^n$ .
5. (i)  $3 \times 3 = 9$   
(ii)  $2 \times \binom{3}{2} = 2 \times 3 = 6$   
(iii) 0, as the sets have differing cardinalities.  
(iv)  $2 \times 2 \times 2 = 8$   
(v)  $2 \times 2 \times 2 - \{\text{all elements map to } 2\} - \{\text{all elements map to } 3\} = 8 - 2 = 6$
6. (a) let  $n = 1$ :

$$\sum_{1 \leq i \leq 1} (2i - 1) = 1$$

$$1^2 = 1$$

$\implies$  true for  $n = 1$ .  
assume  $\exists n = k$  such that:

$$\sum_{1 \leq i \leq k} (2i - 1) = k^2$$

$$\frac{2(k)(k+1)}{2} - k = k^2$$

let  $n = k + 1$ :

$$\begin{aligned} \sum_{1 \leq i \leq k+1} (2i - 1) &= \sum_{1 \leq i \leq k} (2i - 1) + 2(k + 1) - 1 \\ &= k^2 + 2k + 2 - 1 \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$

$\implies$  true for  $n = k + 1$ .  
 $\therefore$  as true for  $n = 1$  and true for  $n = k \implies$  true for  $n = k + 1 \implies$   
 $\sum_{1 \leq i \leq n} (2i - 1) = n^2 \ \forall n \in \mathbb{Z}^+$ .  
 (b) let  $n = 1$ :

$$\begin{aligned}
 \sum_{1 \leq i \leq 1} i^2 &= 1^2 = 1 \\
 \frac{1}{6}(1)(2)(3) &= 1
 \end{aligned}$$

$\implies$  true for  $n = 1$ .  
 assume  $\exists n = k$  s.t:

$$\sum_{1 \leq i \leq k} i^2 = \frac{1}{6}k(k+1)(2k+1)$$

consider  $n = k + 1$ :

$$\begin{aligned}
 \sum_{1 \leq i \leq k+1} i^2 &= \sum_{1 \leq i \leq k} i^2 + (k+1)^2 \\
 &= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2 \\
 &= \frac{1}{6}k(k+1)(2k+1) + k^2 + 2k + 1 \\
 &= \frac{1}{6}(k^2 + k)(2k+1) + k^2 + 2k + 1 \\
 &= \frac{1}{6}(2k^2 + 2k^2 + k^2 + k) + k^2 + 2k + 1 \\
 &= \frac{1}{6}(2k^3 + 3k^2 + k + 6k^2 + 12k + 6) \\
 &= \frac{1}{6}(2k^3 + 9k^2 + 13k + 6) \\
 &= \frac{1}{6}(k+1)(2k^2 + 7k + 6) \\
 &= \frac{1}{6}(k+1)(k+2)(2k+3) \\
 &= \frac{1}{6}(k+1)((k+1)+1)(2(k+1)+1)
 \end{aligned}$$

$\implies$  true for  $n = k + 1$ .  
 $\therefore$  as true for  $n = 1$  and true for  $n = k \implies$  true for  $n = k + 1 \implies$   
 $\sum_{1 \leq i \leq n} i^2 = \frac{1}{6}n(n+1)(2n+1) \ \forall n \in \mathbb{Z}^+$ .

(c)

$$\begin{aligned}
\sum_{3 \leq i \leq n-2} i^2 &= \sum_{1 \leq i \leq n} - \sum_{1 \leq i \leq 2} -n^2 - (n-1)^2 \\
&= \frac{1}{6}n(n+1)(2n+1) - \frac{1}{6}(2)(3)(5) - n^2 - n^2 + 2n - 1 \\
&= \frac{1}{6}(2n^3 + 3n^2 + n) - 5 - 2n^2 + 2n - 1 \\
&= \frac{1}{6}(2n^3 + 3n^2 + n - 30 - 12n^2 + 12n - 6) \\
&= \frac{1}{6}(2n^3 - 9n^2 + 13n - 36)
\end{aligned}$$

(d) let  $n = 1$ :

$$\begin{aligned}
7^1 - 1 &= 6 \\
6 \div 6 &= 1 \\
\implies 6 &\mid 6
\end{aligned}$$

$\therefore$  true for  $n = 1$ .  
assume  $\exists n = k$  s.t:

$$7^k - 1 \mid 6$$

consider  $n = k + 1$ :

$$\begin{aligned}
7^{k+1} - 1 &= 7(7^k) - 1 \\
&= 7(7^k) - 7 + 6 \\
&= 7(7^k - 1) + 6
\end{aligned}$$

as  $7^k - 1 \mid 6$  and  $6 \mid 6 \implies$  true for  $n = k + 1$ .

$\therefore$  as true for  $n = 1$  and true for  $n = k \implies$  true for  $n = k + 1 \implies$   
 $7^n - 1 \mid 6 \forall n \in \mathbb{Z}^+$ .

(e) let  $n = 3$ :

$$\begin{aligned}
2(3) + 1 &= 7 \\
2^3 &= 8 \\
7 &\leq 8
\end{aligned}$$

$\implies$  true for  $n = 1$   
assume  $\exists n = k$  s.t:

$$2k + 1 \leq 2^k$$

consider  $n = k + 1$ :

$$2(k + 1) + 1 = 2k + 2 + 1 = 2k + 3$$

$$\text{recall } 2k + 1 \leq 2^k$$

$$\implies 2k + 3 \leq 2^k + 2$$

$$2^{k+1} = 2(2^k) = 2^k + 2^k$$

$$\implies 2^k + 2^k \geq 2^k + 2$$

$$\implies 2k + 3 \leq 2^k + 2^k$$

$$2k + 3 \leq 2^{k+1}$$

$$\therefore 2(k + 1) + 1 \leq 2^{k+1}$$

$$\implies \text{true for } n = k + 1.$$

$$\therefore \text{as true for } n = 1 \text{ and true for } n = k \implies \text{true for } n = k + 1 \implies 2n + 1 \leq 2^n \forall n \in \mathbb{Z}^+, n \geq 3.$$

7. (i) reflexive: for any  $x \in \mathbb{Z}$ ,  $x + x = 2x$ .  $2x$  is even  $\implies$  not reflexive.

symmetric: as  $x + y = y + x$  ( $+$  is symmetric), the relation is symmetric  $\forall x, y \in \mathbb{Z}$ .

transitive:  $2 + 3 = 5$ , which is odd, and  $3 + 4 = 7$  which is odd.  $2 + 4 = 6$  which is not odd.  $\implies$  not transitive.

$\therefore$  symmetric.

- (ii) reflexive: for any  $x \in \mathbb{Z}$ ,  $x + x = 2x$ .  $2x$  is even  $\implies$  reflexive.

symmetric: as  $x + y = y + x$  ( $+$  is symmetric), the relation is symmetric  $\forall x, y \in \mathbb{Z}$ .

transitive:  $2 + 4 = 6$ , which is even and  $4 + 8 = 12$  which is even.  $2 + 8 = 10$  which is even.  $\implies$  transitive.

$\therefore$  reflexive, symmetric, and transitive.

- (iii) reflexive: for any  $x \in \mathbb{Z}$ ,  $xx = x^2$ .

if  $x$  is even:

$$(2k)^2 = 4k^2$$

$$\implies \text{even.}$$

if  $x$  is odd:

$$(2k + 1)^2 = 4k^2 + 4k + 1$$

$$\implies \text{odd.}$$

$$\implies \text{not reflexive.}$$

symmetric: as  $xy = yx$  ( $\times$  is symmetric), the relation is symmetric  $\forall x, y \in \mathbb{Z}$ .

transitive:  $3(5) = 15$ , which is odd, and  $5(7) = 35$ .  $3(7) = 21$ , which is odd.  $\implies$  transitive.

$\therefore$  symmetric and transitive.

(iv) reflexive: for any  $x \in \mathbb{Z}$ ,  $x + x(x) = x(x + 1)$ .

if  $x$  is even:

$$(2k)(2k + 1) = 4k^2 + 2k$$

$\implies$  even.

if  $x$  is odd:

$$(2k + 1)(2k + 2) = 4k^2 + 6k + 2$$

$\implies$  even.

$\implies$  reflexive.

symmetric: if  $x$  is even,  $xy$  is even.

$y + (\text{even})$  is only even if  $y$  is even.

$\therefore$  if  $y$  is odd, the relation is not symmetric.  $\implies$  not symmetric  $\forall x, y \in \mathbb{Z}$ .

transitive:  $2 + 2(3) = 8$ , which is even, and  $3 + 3(5) = 18$ , which is even.  $2 + 2(5) = 12$ , which is even.  $\implies$  transitive.

$\therefore$  reflexive and transitive.

8. (a) reflexive:  $\subset$  is reflexive, as for any set  $A$ ,  $A \subset A$  as every element from  $A$  is also in  $A$ .

antisymmetric: if  $A \subset B$  and  $B \subset A$ , then every element in  $B$  is in  $A$ , and every element in  $A$  is also in  $B$ , so  $A = B$ .

transitive:  $\{b\} \subset \{a, b\}$ , and  $\{a, b\} \subset \{a, b, c\}$  and  $\{b\} \subset \{a, b, c\}$ .

$\{c\} \subset \{a, c\}$ ,  $\{a, c\} \subset \{a, b, c\}$  and  $\{c\} \subset \{a, b, c\}$ .

$\implies C$  is transitive on  $A$ .

$\therefore \subset$  is partial order on  $A$ .

- (b)  $\subset$  is not total order on  $A$ , as  $\{b\} \not\subset \{c\}$  and  $\{c\} \not\subset \{b\}$ , and for a total order relation, every element is related to another in some direction.

- (c)  $\{a, b, c\}$  is a maximal element, as it is only a subset of itself. This is because every other element has a smaller cardinality than it, and so cannot be related to (a subset of) it.

$\{b\}$  and  $\{c\}$  are minimal elements. This is because they both have a

cardinality of 1, and so the only elements that can be a subset of it also have a cardinality of 1, and  $\{b\} \not\subset \{c\}$  &  $\{c\} \not\subset \{b\}$ .